

THE TWO MOST SYMMETRIC FLAT TORI ARE VERTEX-MINIMAL PAPER TORI

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ABSTRACT. A paper torus is a polyhedral torus in $\mathbb{R}^3$ built from flat triangles. Schwartz proved that a paper torus needs at least eight vertices and that eight can be achieved. Doyle and Schwartz then realized almost every shape of flat torus with eight vertices, but their construction degenerates along the rectangular tori and the rhombic tori of aspect ratio at least $\sqrt{3}$, and the two most symmetric flat tori, the square torus and the hexagonal torus, sit at the two corners of what is left out. We build embedded paper tori with eight vertices whose intrinsic geometries are exactly those of the square torus and of the hexagonal torus, so both are vertex-minimal. The construction is elementary and the arithmetic is exact. Near each of the two tori the proof produces a smooth seventeen dimensional manifold of exactly flat configurations on which the modulus is a submersion, so eight-vertex paper tori also realize every nearby shape.

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1. INTRODUCTION

Take a flat sheet of paper, and fold and glue it into a torus. The intrinsic geometry of the sheet never changes, and the result is a *flat torus*, a surface with zero Gaussian curvature everywhere, realized in three-space as a polyhedral surface. Such *paper tori* were first constructed by Burago and Zalgaller [1], who later proved that every isometry class of flat torus is realized, though only by constructions of enormous size [2] (see [8]). Explicit constructions followed, from Brehm's diplotori [3] to origami tori [4] and the diplotorus family of Arnoux, Lelièvre, and Málaga [5]. Quintanar embedded the square torus explicitly [6], and Lazarus and Talerie built a single universal triangulation, with 2434 faces, admitting every flat torus [8]. An earlier 8-vertex flat square torus of Gott and Vanderbei [7] is not embedded. The modern question is how few vertices a paper torus can have. Tugayé found one with 9 [9], and Schwartz then settled the

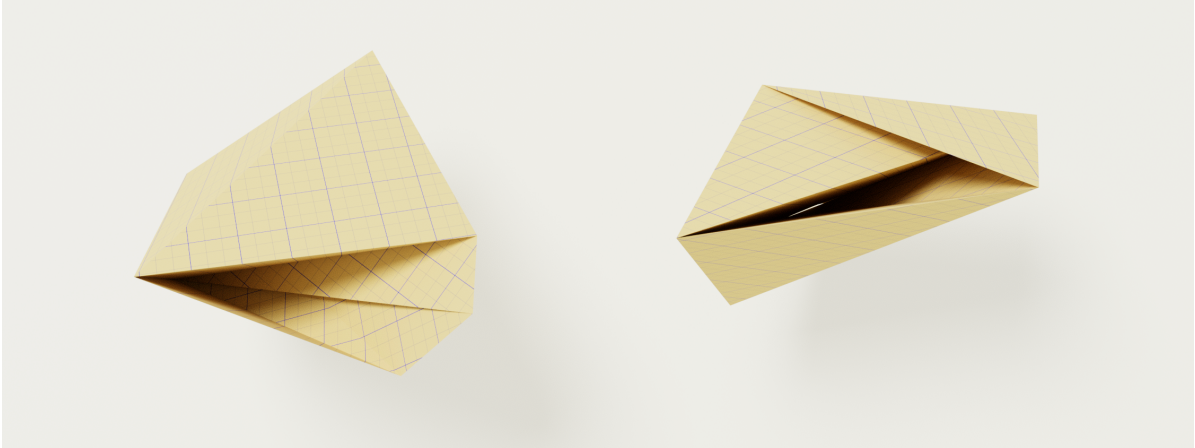


FIGURE 1. Two inflated tori numerically approximating paper tori, the square one left and the hexagonal one right, not the nearly planar objects of the proofs, see Appendix B.

question, proving that no paper torus has 7 vertices and that 8-vertex paper tori exist [10]. Figure 1 gives a sense of what a paper torus with only eight vertices can look like.

Which shapes of flat torus arise with eight vertices? The shape is captured by a single point $\hat{\tau}$ in the classical modular domain, the *reduced modulus* (Section 2). Doyle and Schwartz proved near universality [11]. The pup tent [10] carries an order-2 rotational symmetry, its moduli sweep out the interior of the bi-cusped fundamental domain of [11], in the coordinates of Section 2 the $\hat{\tau}$ with $0 < |\operatorname{Re} \hat{\tau}| < \frac{1}{2}$, the open arc of $|\hat{\tau}| = 1$ between the orbifold points included, and in particular every flat torus *without reflection symmetry* is an 8-vertex paper torus. Excluded are exactly the rectangular tori $\{iY \mid Y \geq 1\}$ and the rhombic tori of aspect ratio at least $\sqrt{3}$, the set $\{\pm \frac{1}{2} + iy \mid y \geq \frac{\sqrt{3}}{2}\}$. Doyle and Schwartz ask whether these excluded shapes can be realized if one drops the symmetry of the pup tent.

This paper answers that question for the two most symmetric points, the square torus $\hat{\tau} = i$, realized on the same triangulation as the pup tents, and the hexagonal torus $\hat{\tau} = \rho = e^{i\pi/3}$, realized on a second triangulation. They are the two orbifold points of the moduli space, at the corners of the excluded locus. Doyle and Schwartz come arbitrarily close, their collapsing families containing 8-vertex paper tori whose moduli converge to these two points. But the tori themselves degenerate, their vertices colliding in the limit. The square torus had been embedded polyhedrally before, by Quintanar with 40 vertices [6]. Our result is that eight suffice, the fewest possible, by Schwartz's lower bound [10].

Theorem 1. *Every square flat torus is realized by an 8-vertex paper torus.*

Theorem 2. *Every hexagonal flat torus is realized by an 8-vertex paper torus.*

Both tori are vertex-minimal, by Schwartz's bound. Since all square tori are equivalent via scaling, it is enough to prove Theorem 1 for a single square torus, and the same applies to Theorem 2.

Our construction is almost completely explicit. For the square torus we produce a path q_t of embedded paper tori for $0 < t < \varepsilon$. The limit q_0 is an explicit planar configuration in $\mathbb{R}^2 \times \{0\}$, its eight vertices the rational points of Table 1, and as t varies the points move continuously. We call q_0 the *base*. For each vertex v ,

$$q_t(v) = q_0(v) + (x_v(t), y_v(t), \zeta_v t),$$

where x_v and y_v are continuous functions vanishing at 0 and ζ_v is an explicit rational value. We call $\zeta_0, \dots, \zeta_7$ the lift parameters.

The hexagonal case is the same, except that the vertices of q_0 and the lift parameters lie in $\mathbb{Q}(\sqrt{3})$ and the triangulation is T' rather than the valence-regular T . For both cases we list the vertex coordinates in Table 1 and the lift parameters in Section 5. The continuous functions are not explicit. They are obtained as a byproduct of a call to the implicit function theorem.

Both theorems come with a stronger statement attached. Near the two new tori the flat configurations form a smooth seventeen dimensional manifold on which the modulus is a submersion (Theorem 3), so every nearby modulus is realized by an embedded configuration as well, and eight-vertex paper tori realize initial segments of both excluded rays (Corollary 1).

The tori of both theorems are nearly planar. Appendix B records an inflated torus for each (Figure 1), found numerically and exactly embedded, with printable nets to cut out and fold.

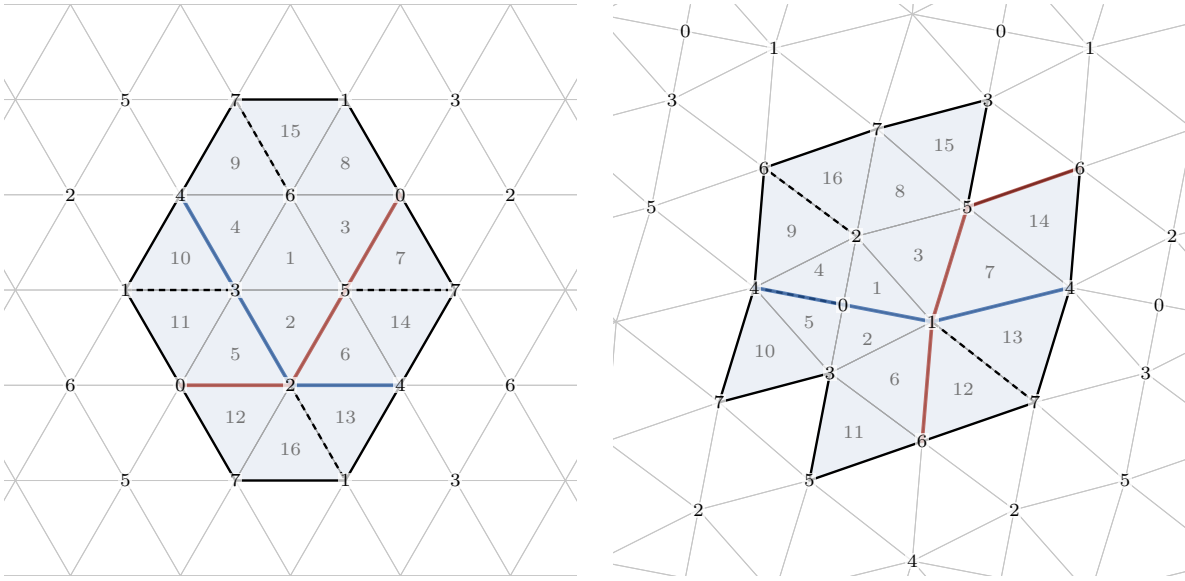


FIGURE 2. The two combinatorial triangulations through their universal covers. On the left T , whose cover is the regular $\{3, 6\}$ triangulation of the plane. On the right the second triangulation T' , of degrees $(4, 6, 6, 6, 6, 6, 7, 7)$. In each panel the shaded region is the net of the unfolding of Section 4, 16 triangles on 8 vertices labeled 0 through 7.

Both theorems are proved the same way. We begin from a planar configuration of eight points, folded flat, whose eight vertices are points of $\mathbb{Q}^2$ for the square torus and of $\mathbb{Q}(\sqrt{3})^2$ for the hexagonal one, and whose flatness and modulus are exact identities in the field. Raising the eight vertices along one fixed rational direction pulls the overlapping sheets apart (Section 5), and an implicit function theorem argument then corrects the fold back to exact flatness and exact modulus (Section 6). The correction happens within a neighborhood which is strictly contained in a neighborhood of embedded configurations, proving the theorem.

Flatness of a configuration is usually stated through its cone angles, which are sums of arccosines. We see it instead by unfolding (Section 4). Laying the sixteen faces into the plane one after another using a fixed order produces eighteen boundary edges paired up into nine gluings. From this perspective, a configuration is flat if and only if all nine gluings are translations. For our two bases several of the nine translations even vanish, four for the square

torus and three for the hexagonal one, so the nets close partway back up along the dashed edges of Figure 2. The same picture measures the modulus, as the ratio of the two translations the unfolding picks up along a fixed pair of loops. For any given fold the unfolding is a composition of reflections in lines through pairs of vertices, so all of this is arithmetic in the field of the base, and the correction differentiates cone angles, whose derivatives are rational as well once evaluated at the fold. All checks are short exact computations, reproducible in any computer algebra system.

Acknowledgements. The results were found using data that came out of code written jointly with Steve Trettel in early June 2026, as a gift for Richard Schwartz’s birthday conference, made in the name of Anna Wienhard’s group at MPI MiS Leipzig. It grew out of many conversations with Steve, from which I learned a great deal, and I am grateful to him for his generosity and insight throughout. I thank Richard Schwartz for many conversations and insights, and for his encouragement to write up this result. I thank Marit Bobb and Clara Winne, who folded the tori of Appendix B from paper when I could not. [More official stuff that I don’t know about.]

2. FLAT TORI, PAPER TORI, AND THE TWO CORNERS

A *flat torus* is the quotient $\mathbb{C}/\Lambda$ of the plane by a lattice $\Lambda = \mathbb{Z}v_1 \oplus \mathbb{Z}v_2$. Rotating and scaling the plane changes (v_1, v_2) to $(\alpha v_1, \alpha v_2)$ without changing the intrinsic shape, so the shape is captured by the ratio $\tau = v_2/v_1$, which lies in the upper half plane $\mathbb{H}$ when the basis is ordered positively, as we always require, and changing the basis of Λ moves τ by an element of the modular group $\mathrm{PSL}(2, \mathbb{Z})$. Every orbit meets the closed fundamental domain $\mathcal{D} = \{|\tau| \geq 1, |\mathrm{Re} \tau| \leq \frac{1}{2}\}$ (Figure 3), and its representative there is the *reduced modulus* $\hat{\tau}$ (see [14, §12.2]). On the boundary of $\mathcal{D}$ the representative is not unique, the gluing $\tau \mapsto \tau + 1$ identifying the two vertical sides and $\tau \mapsto -1/\tau$ the two halves of the arc, and we always take the representative with $\mathrm{Re} \hat{\tau} \geq 0$. The orbit of i meets $\mathcal{D}$ in the single point i , so for the square torus the representative is unambiguous. The orbit of $\rho = e^{i\pi/3}$ meets $\mathcal{D}$ in the two points $\pm \frac{1}{2} + i\frac{\sqrt{3}}{2}$, and the convention picks $\hat{\tau} = \rho$. The tori with an orientation-reversing symmetry are exactly the rectangular tori $\hat{\tau} = iY$, $Y \geq 1$, and the *rhombic* tori, those whose lattice is generated by two vectors of equal length, in $\mathcal{D}$ the arc $|\hat{\tau}| = 1$ together with the sides $\mathrm{Re} \hat{\tau} = \pm \frac{1}{2}$. The *aspect ratio* of a rhombic torus is the ratio of the diagonals of its rhombus, $2y$ at $\hat{\tau} = \pm \frac{1}{2} + iy$ and below $\sqrt{3}$ on the arc. The square torus is $\hat{\tau} = i$, its lattice having a rotational symmetry of order four, and the hexagonal torus is $\hat{\tau} = \rho$, of order six.

Fix a triangulation T of the torus with vertex set V . Every face is a triangle and every edge lies in two faces, so $2|E| = 3|F|$, and Euler’s formula $|V| - |E| + |F| = 0$ gives $|E| = 3|V|$ and $|F| = 2|V|$. A triangulation is simplicial, so $|E| \leq |V|(|V| - 1)/2$, and $3|V| \leq |V|(|V| - 1)/2$ forces $|V| \geq 7$. For $|V| = 8$, the case of this paper, T has 24 edges and 16 faces, and up to combinatorial isomorphism there are exactly seven such T [12, 13]. A *configuration* is a map $q: V \rightarrow \mathbb{R}^3$, inducing a piecewise linear map $|q|$ of T into space, affine on each face. The configuration is *flat* if the cone angle at every vertex, the sum of the incident triangle angles measured in $\mathbb{R}^3$, equals 2π . Write δ_v for the *deficit* at v , that is, 2π minus the cone angle there. The sixteen triangles carry total angle 16π , so

$$\sum_{v \in V} \delta_v = 8 \cdot 2\pi - 16\pi = 0$$

for every configuration with nondegenerate faces, and flatness is the vanishing of any seven of the eight deficits. The intrinsic metric then has no cone points and the surface is a flat torus with a well-defined modulus. We call an embedded polyhedral surface in $\mathbb{R}^3$ whose intrinsic

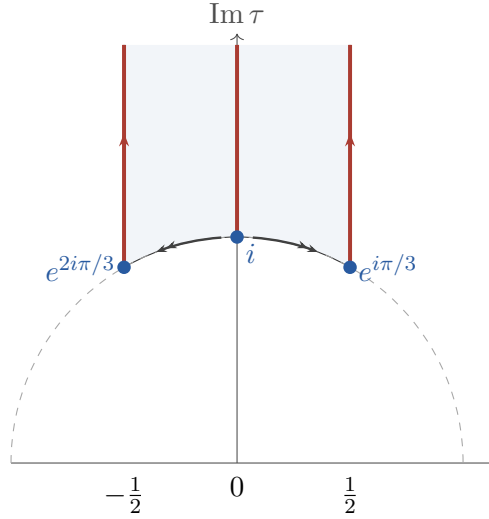


FIGURE 3. The fundamental domain $\mathcal{D}$ of the modular group, the arrows being the identifications that fold it into the modular surface. The pup tents of [11] realize every modulus off the rectangular line $\operatorname{Re} \hat{\tau} = 0$ and the rhombic sides $\operatorname{Re} \hat{\tau} = \pm \frac{1}{2}$. The open arc between i and ρ is among the moduli they realize. This paper realizes the two orbifold points i and ρ at its corners.

metric is a flat torus a *paper torus*, the notion of [10, 11]. Equivalently, a paper torus is the image of a piecewise affine isometric embedding $\phi: T \rightarrow \mathbb{R}^3$ of a flat torus T , the faces of the triangulation going to the pieces on which ϕ is affine. We say that T is *realized* by that paper torus. A configuration that is both flat and injective is one, with eight vertices.

2.1. The geometry of the problem. Fix one of the seven triangulations T , together with a marking, an ordered pair of edge loops whose classes form a basis of $H_1(T; \mathbb{Z})$ (Section 4). The configurations form the space

$$\mathcal{C} = (\mathbb{R}^3)^V \cong \mathbb{R}^{24},$$

and we single out three subsets of it. The set $\mathcal{C}^\circ$ of configurations with nondegenerate faces is open. Inside it, the set $\mathcal{E}$ of embedded configurations is open as well, since embeddedness is an open condition on a configuration. The flat configurations form the *flat locus* $\mathcal{F} \subset \mathcal{C}^\circ$, cut out by the vanishing of seven of the eight deficits, the eighth vanishing with them by the angle count above. The universal cover of the intrinsic metric of a point of $\mathcal{F}$ is the Euclidean plane, each class of $H_1(T; \mathbb{Z})$ acting on it by a translation, and this identifies $H_1(T; \mathbb{Z})$ with the lattice Λ of a flat torus $\mathbb{R}^2/\Lambda$, up to rotation and scaling. The marking becomes a basis (v_1, v_2) of Λ , and we write

$$\tau: \mathcal{F} \longrightarrow \mathbb{H}, \quad \tau = v_2/v_1,$$

the ratio being unchanged by the rotation and the scaling and lying in $\mathbb{H}$ for a positively ordered basis, as above. Section 4 computes τ exactly, by unfolding. The representative of τ inside the fundamental domain $\mathcal{D}$ gives the reduced modulus $\hat{\tau}$.

A point of $\mathcal{E} \cap \mathcal{F}$ is a paper torus with eight vertices. Schwartz's theorem [10] says that $\mathcal{E} \cap \mathcal{F}$ is nonempty on the valence-regular triangulation, and near universality [11] says that every reduced modulus off the two excluded rays, the rectangular ray $\hat{\tau} = iY$ with $Y \geq 1$ and the rhombic ray $\hat{\tau} = \frac{1}{2} + iy$ with $y \geq \frac{\sqrt{3}}{2}$, can be achieved. Theorems 1 and 2 produce points of

$\mathcal{E} \cap \mathcal{F}$ with $\hat{\tau} = i$ and $\hat{\tau} = \rho$, one on each of the two triangulations of Section 3, and Theorem 3 describes $\mathcal{F}$ and the fibers of τ near them.

The group of orientation-preserving similarities of $\mathbb{R}^3$ has dimension seven, three translations, three rotations and one scaling. It acts on $\mathcal{C}$ preserving $\mathcal{C}^\circ$, $\mathcal{E}$ and $\mathcal{F}$, and it fixes τ , a similarity changing no cone angle and no period ratio. Quotienting by it is the same as restricting to unit area and quotienting by orientation-preserving isometries, every similarity orbit meeting the unit-area configurations in exactly one isometry orbit. Every dimension below is stated in $\mathcal{C}$ and carries the seven orbit directions with it, and the fifteen of Theorem 3 is $7 + 8$, seven along the similarity orbit and eight that change the polyhedron.

3. THE TWO FOLDED TORI

The construction starts from two planar objects, one for the square torus and one for the hexagonal torus. Each is a configuration folded flat into the plane with its eight vertices at explicit coordinates of a number field ($\mathbb{Q}$ and $\mathbb{Q}(\sqrt{3})$ respectively), its sixteen triangles nondegenerate. That each one is in fact a flat torus, of modulus exactly i in one case and exactly ρ in the other, is settled in Section 4 by unfolding it.

Throughout, a triangulation is presented by its face list, with vertex set V , edge set E , and face set F , faces written as positively ordered triples. One can see in Figure 2 that the two printed lists triangulate the torus, each as a net of sixteen triangles with its boundary glued in pairs, and Figure 5 shows the same with the true shapes of the faces.

3.1. The square base. The triangulation T has vertex set $V = \{0, \dots, 7\}$ and the sixteen faces

$$\begin{aligned} &356, 325, 364, 302, 341, 310, 506, 524, \\ &547, 570, 674, 601, 617, 214, 207, 271, \end{aligned}$$

written as positive boundary cycles, in the fixed order used by every table and script, its 24 edges being the pairs of vertices that occur in them. Among the seven combinatorial types of 8-vertex torus triangulation [12, 13] it is the one in which every vertex has valence 6, the type of Schwartz's pup tents [10].

The square base is the planar configuration $Q^0: V \rightarrow \mathbb{Q}^2$ of Table 1, read as a configuration in $\mathbb{R}^3$ with all z coordinates zero, and drawn on the left of Figure 4. Its sixteen faces are nondegenerate, with orientation signs

$$+ + - + + - - + - + - + + - - -$$

in the face order above, so eight of them are folded over. The planar vertices of both bases satisfy exact geometric relations. They play no role in the proof and are listed in Section 7.1.

3.2. The hexagonal base. The triangulation T' has vertex set $\{0, \dots, 7\}$, the sixteen faces

$$\begin{aligned} &012, 031, 024, 043, 152, 136, 145, 174, \\ &167, 264, 257, 276, 347, 356, 375, 465, \end{aligned}$$

written as positive boundary cycles. The vertex degrees are $(4, 6, 6, 6, 6, 6, 7, 7)$, the degree-7 vertices being 1 and 4 and the degree-4 vertex 0. Its vertex cycles are now of lengths $(4, 7, 6, 6, 7, 6, 6, 6)$ in vertex order.

The hexagonal base is the planar configuration P^0 of Table 1, again with heights zero, and drawn on the right of Figure 4. Every number in it lies in $\mathbb{Q}(\sqrt{3})$, written $a + b\sqrt{3}$ with $a, b \in \mathbb{Q}$, and every claim we make about it is decided by exact arithmetic in this field. Its sixteen faces are nondegenerate, with orientation signs

$$- + + - - - - + - + + + - +$$

in the face order above, so here too eight are folded over.

v	Q_v^0	P_v^0
0	$(0, 0)$	$(0, 0)$
1	$(\frac{73}{60}, \frac{11}{10})$	$\frac{1}{8073677900} (896842303 + 729553300\sqrt{3}, 1029317275 + 568256878\sqrt{3})$
2	$(\frac{1}{2}, 0)$	$\frac{1}{47100} (-2625 + 7310\sqrt{3}, 8925 - 24854\sqrt{3})$
3	$(-\frac{15977}{50320}, \frac{42471}{50320})$	$\frac{1}{10525} (3789, 595 - 4713\sqrt{3})$
4	$(1, 0)$	$(1, 0)$
5	$(\frac{35937}{25160}, \frac{27357}{25160})$	$\frac{1}{200} (64 + 25\sqrt{3}, 75 - 136\sqrt{3})$
6	$(0, \frac{33}{20})$	$\frac{1}{200} (-36 + 25\sqrt{3}, 75 - 36\sqrt{3})$
7	$(1, \frac{33}{20})$	$\frac{1}{109900} (279568 - 117825\sqrt{3}, 264900 - 143557\sqrt{3})$

TABLE 1. The two folded bases, the eight planar vertex positions of each, Q^0 exact in $\mathbb{Q}$ and P^0 exact in $\mathbb{Q}(\sqrt{3})$, the hexagonal entries each scaled by one common denominator.

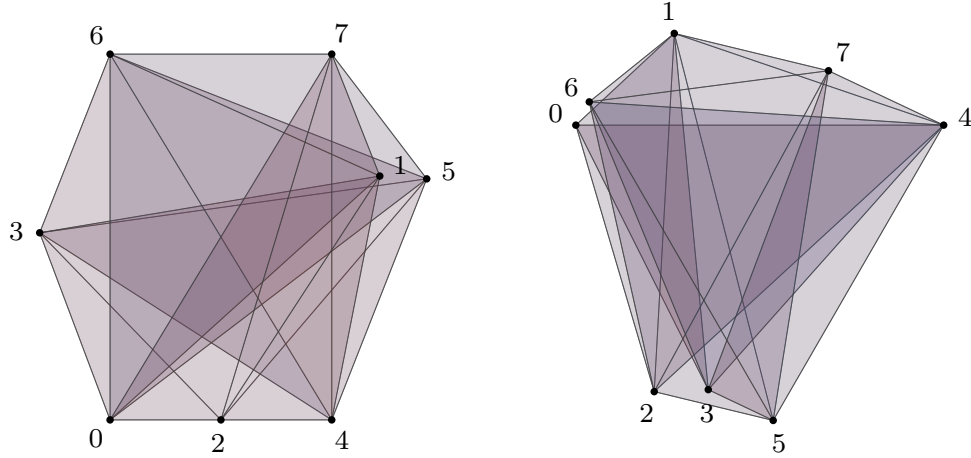


FIGURE 4. The two folded bases. On the left the square base Q^0 , on the right the hexagonal base P^0 , whose fold is the tighter of the two.

Remark 1. Neither base is ad hoc. The square one, the column Q^0 of Table 1, is the member at $Y = 1$ of a four-parameter rational family of folded flat tori of modulus iY that covers the entire rectangular moduli line, and the hexagonal one, the column P^0 , is the corner member of a second such family, sweeping the rhombic line $\hat{\tau} = \frac{1}{2} + ih$, $h \geq \frac{\sqrt{3}}{2}$, at the endpoint where its coordinates are in $\mathbb{Q}(\sqrt{3})$. Those families, and the arguments that prove theorems analogous to Theorems 1 and 2 for both lines, are the subject of a sequel paper in preparation, see Section 7. The present paper is self-contained and consumes Table 1 and the two directions of Section 5, nothing else.

4. THE UNFOLDING

We will work with the unfolded configuration and use it to show flatness and to read off the modulus. For this we fix, once and for all, the order in which the sixteen faces are unfolded, by the same rule on both triangulations. The first face of the printed list goes down first.

Whenever a face is down, its three neighbours go down next, in the order the shared edges appear in its boundary cycle, skipping faces that are already placed down. This is the breadth first order on the dual graph rooted at the first face, the faces of each level laid down in the order they were reached, which pins the order completely. It glues fifteen of the 24 edges, and the resulting net, sixteen faces joined along the fifteen tree edges, is shaded in Figure 2 for both triangulations, drawn in the universal cover. The nine remaining edges are used to determine if the configuration is flat. For a flat configuration, we can read off the modulus from the unfolding, as the ratio of the two translations the unfolding picks up along a fixed pair of loops generating the first homology. The loops are also indicated in Figure 2, and the translations are the sums of the unfolded edge vectors along them.

A configuration q whose sixteen faces are nondegenerate carries a metric. Each face is given the metric of its spatial triangle, and the two triangles at an edge have the same length there, so they glue up by an isometry. The result is a Euclidean cone metric on T , flat away from the eight vertices, whose cone angle at v is the sum of the triangle angles at v . Everything computed from q below is computed from the 24 squared edge lengths

$$L_{ab}(q) = |q_a - q_b|^2,$$

since the angles of a Euclidean triangle are determined by its squared side lengths and, as we now describe, so is the unfolding.

The unfolding assigns to each face f and each vertex v of f a point $u_f(v)$ of the plane. The root face goes down with $u(a) = (0, 0)$ and $u(b) = (1, 0)$, each further face across its shared edge. Every $u_f(v)$ is a rational function of the squared edge lengths and the doubled areas S_f , so a square root enters only through S_f . Here $S_f = 2 \text{Area}(f) = \sqrt{L_{ab}L_{ac} - \langle ab, ac \rangle^2}$ and $\langle ab, ac \rangle = \frac{1}{2}(L_{ab} + L_{ac} - L_{bc})$. Every unfolded face is a positively oriented copy of its spatial face at the scale of the root edge. At a planar configuration each radicand is a perfect square, by Lagrange's identity $|u|^2|v|^2 - (u \cdot v)^2 = (u_1v_2 - u_2v_1)^2$ for the two plane vectors spanning the face, so on the two bases the unfolding is exact, the fourteen and fifteen unfolded vertices of Tables 2 and 3, in $\mathbb{Q}$ and in $\mathbb{Q}(\sqrt{3})$. And it is smooth wherever the sixteen faces are nondegenerate, which is what Section 6 differentiates. Figure 5 draws both nets, with the same face order as in Figure 2.

The forty-eight placed corners fall on fourteen points of the plane on the square side and on fifteen on the hexagonal one, the corners of the two nets, and Tables 2 and 3 give them exactly, each point named $u_f(v)$ by the vertex v it is a copy of and the face f that places it first.

After unfolding we are left with nine non-glued edges for each configuration. These are the boundary edges of the fundamental domain and the interior dashed edges in Figure 2. To show that a configuration is flat, it suffices to show that for each such edge $\{p, q\}$, shared by two faces f and g , the unfolded edge vectors agree, i.e. that

$$u_f(p) - u_f(q) = u_g(p) - u_g(q),$$

which makes each of the nine gluings a translation, every unfolded face being positively oriented. Indeed, going once around a vertex composes the transitions between the consecutive incident faces, the identity across a tree edge and a gluing across a non-tree edge, so the composition is a product of gluings and their inverses, and it is the holonomy of the development around the vertex. Its linear part is the rotation by the cone angle there. If every gluing is a translation, every such composition is a translation, its linear part is the identity. Hence each cone angle is a positive multiple of 2π and the sixteen triangles carry total angle 16π between the eight

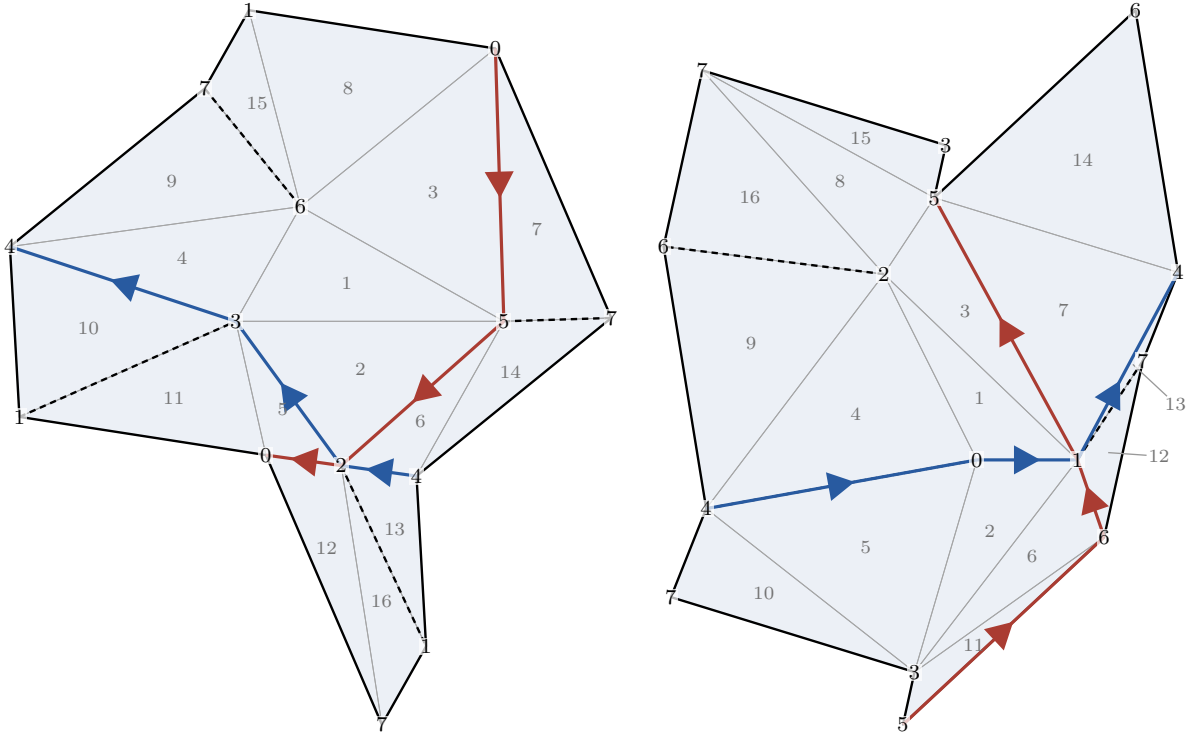


FIGURE 5. The two nets, sixteen faces unfolded along the dual tree, the square base left and the hexagonal base right. Thin interior segments are the fifteen tree edges, the heavy outline is the free boundary, and dashed segments are non-tree edges along which the net closes back up, their deck translations zero. The blue and red arrows are the unfolded edge vectors of the two marking loops, each triple summing to its period.

point	position
$u_{356}(3)$	$(0, 0)$
$u_{356}(5)$	$(1, 0)$
$u_{356}(6)$	$\frac{1}{6254125} (1510441, 2676762)$
$u_{325}(2)$	$\frac{1}{6254125} (2459423, -3366264)$
$u_{506}(0)$	$\frac{1}{6254125} (6055927, 6365964)$
$u_{364}(4)$	$\frac{1}{250165} (-210837, 69696)$
$u_{302}(0)$	$\frac{1}{6254125} (702403, -3121404)$
$u_{524}(4)$	$\frac{1}{6254125} (4216443, -3611124)$
$u_{570}(7)$	$\frac{1}{6254125} (8761929, 78078)$
$u_{601}(1)$	$\frac{1}{18762375} (915847, 21774654)$
$u_{674}(7)$	$\frac{1}{6254125} (-725439, 5431602)$
$u_{341}(1)$	$\frac{1}{750495} (-605789, -267498)$
$u_{207}(7)$	$\frac{1}{1250825} (681681, -1881858)$
$u_{214}(1)$	$\frac{1}{18762375} (13317379, -22748022)$

TABLE 2. The square net of Figure 5. Its sixteen faces put their forty-eight corners on fourteen points of the plane, each named here by the face that places it first and listed in the order the unfolding places them, exact in $\mathbb{Q}$ and at the scale of the root edge.

point		position
$u_{012}(0)$	x	0
	y	0
$u_{012}(1)$	x	1
	y	0
$u_{012}(2)$	x	$\frac{1}{555694913304951} (2515566967447176 - 1748881409442255\sqrt{3})$
	y	$\frac{1}{555694913304951} (6921139243529985 - 3403655728740217\sqrt{3})$
$u_{031}(3)$	x	$\frac{1}{2118893} (-6578800 + 3039430\sqrt{3})$
	y	$\frac{1}{2118893} (-22367920 + 10334062\sqrt{3})$
$u_{152}(5)$	x	$\frac{1}{740926551073268} (544497372063943 - 495475141869625\sqrt{3})$
	y	$\frac{1}{740926551073268} (13541673711173875 - 6703740324583475\sqrt{3})$
$u_{024}(4)$	x	$\frac{1}{185231637768317} (1305770380419850 - 1041620420752000\sqrt{3})$
	y	$\frac{1}{185231637768317} (-1339666698144250 + 721657938141100\sqrt{3})$
$u_{136}(6)$	x	$\frac{1}{740926551073268} (-5077052117214257 + 3472341516107375\sqrt{3})$
	y	$\frac{1}{740926551073268} (-2147368694014125 + 911781885620925\sqrt{3})$
$u_{145}(4)$	x	$\frac{1}{185231637768317} (-1823142417564850 + 1266463823447500\sqrt{3})$
	y	$\frac{1}{185231637768317} (419398228148750 - 42809631023600\sqrt{3})$
$u_{257}(7)$	x	$\frac{1}{1296621464378219} (-33749292045964731 + 17441837192871625\sqrt{3})$
	y	$\frac{1}{1296621464378219} (92012050691116625 - 50224279449101350\sqrt{3})$
$u_{264}(6)$	x	$\frac{1}{740926551073268} (5767578893743343 - 4661956824877625\sqrt{3})$
	y	$\frac{1}{740926551073268} (8183006918596875 - 3817108572019075\sqrt{3})$
$u_{347}(7)$	x	$\frac{1}{1296621464378219} (-30825006729247631 + 15520269580198875\sqrt{3})$
	y	$\frac{1}{1296621464378219} (61620438884996375 - 36597448164078450\sqrt{3})$
$u_{356}(5)$	x	$\frac{1}{740926551073268} (2215517553045143 - 1593513777682625\sqrt{3})$
	y	$\frac{1}{740926551073268} (-3824961606609125 + 1083020409715325\sqrt{3})$
$u_{167}(7)$	x	$\frac{1}{1296621464378219} (-52727396315140531 + 31676859289595375\sqrt{3})$
	y	$\frac{1}{1296621464378219} (73933893369047375 - 41948721148231350\sqrt{3})$
$u_{465}(6)$	x	$\frac{1}{740926551073268} (-6748072298195457 + 4570380151920375\sqrt{3})$
	y	$\frac{1}{740926551073268} (15219266623768875 - 6874978848677875\sqrt{3})$
$u_{375}(3)$	x	$\frac{1}{77982519500461457} (-417997246102752500 + 227430000312057320\sqrt{3})$
	y	$\frac{1}{77982519500461457} (1004622302211424670 - 439227692049635862\sqrt{3})$

TABLE 3. The hexagonal net of Figure 5, its fifteen points read the same way, exact in $\mathbb{Q}(\sqrt{3})$, the two coordinates of a point on consecutive lines and each point scaled by one common denominator.

vertices, so every cone angle is exactly 2π and the configuration is flat.¹ The condition is again a closed expression in the field of the base (so for us either $\mathbb{Q}$ or $\mathbb{Q}(3)$), and on the two folded bases it holds for every pair (a set of simple subtractions of values in Tables 2 and 3).

It remains to identify the two tori. A *marking* is an ordered pair of closed edge loops whose classes form a basis of $H_1(T; \mathbb{Z})$, and we fix $4 \rightarrow 2 \rightarrow 3 \rightarrow 4$ and $5 \rightarrow 2 \rightarrow 0 \rightarrow 5$ on T , and $1 \rightarrow 4 \rightarrow 0 \rightarrow 1$ and $1 \rightarrow 5 \rightarrow 6 \rightarrow 1$ on T' . Each pair consists of simple closed curves crossing exactly once at their shared vertex (Figure 2), with algebraic intersection number ± 1 , so each pair forms a basis.

The *unfolded edge vector* of an edge (a, b) read in a face f that contains it is $u_f(b) - u_f(a)$, well defined at a flat configuration, since any two non-glued edges in the unfolding are related by a translation. The two *periods* v_1, v_2 of a configuration are the sums of the unfolded edge vectors along the two marking loops, in the order the loops are listed above. On both triangulations

¹Alternatively, the nine translations make the unfolding a translation structure on the torus, and a translation structure on the torus has no cone points, apart from removable ones of angle exactly 2π .

that order gives $\det(v_1, v_2) > 0$. Summing the unfolded edge vectors gives, on T ,

$$v_1 = \frac{1}{6254125} (-9487368, 5353524), \quad v_2 = \frac{1}{6254125} (-5353524, -9487368)$$

and therefore $v_2 = iv_1$ (multiplication by i being the quarter turn $(x, y) \mapsto (-y, x)$), so the square base is a square torus. On T' the entries of the two periods lie in $\mathbb{Q}(\sqrt{3})$ with numerators of up to sixteen digits, and $v_2 = \rho v_1$ holds exactly in the same way, so the hexagonal base is a hexagonal torus.

Proposition 1. *The square base Q^0 and the hexagonal base P^0 are flat, with periods satisfying $v_2 = iv_1$ on Q^0 and $v_2 = \rho v_1$ on P^0 , so the intrinsic metric of each is that of the square, respectively the hexagonal, torus up to scaling.*

The proof of the proposition is the discussion so far. Both configurations have nondegenerate faces, with the orientation signs printed in Section 3, so the unfolding applies, and every claim is a finite comparison on printed data, in exact arithmetic in the field of the base (Appendix A): each face of Tables 2 and 3 is congruent to its spatial face, the nine gluings of each base are translations, and the periods, summed along the marking loops, satisfy the two relations. The nine translations make every cone angle exactly 2π , as above.

None of the constructions or formulas of this section used that the base is planar. For a configuration with nondegenerate faces and all cone angles 2π , the same unfolding and marking compute the modulus of Section 2.1, the sum of unfolded edge vectors along a loop being the translation by which its class acts on the universal cover. Section 6 uses this on configurations that are not planar.

5. THE LIFT

We now lift the fold out of the plane, along a fixed rational direction $\zeta \in \mathbb{Q}^V$, one for each base, printed in Proposition 2 below. Throughout this section $Q: V \rightarrow \mathbb{R}^2$ is a planar configuration with nondegenerate faces and $t > 0$. The *lift* of Q places vertex v at $(Q_v, t\zeta_v) \in \mathbb{R}^3$, the ζ_v the lift parameters of Section 1, and the *shadow* of a face $F = (a, b, c)$ is the planar triangle $Q(F)$ that the lifted face sits above. The three values $\zeta_a, \zeta_b, \zeta_c$, prescribed at the corners Q_a, Q_b, Q_c , which are not collinear, determine a unique affine function

$$h_F: \mathbb{R}^2 \rightarrow \mathbb{R}.$$

More concretely, $h_F(x) = \langle m, x \rangle + c$ with $m \in \mathbb{R}^2$ and $c \in \mathbb{R}$. The three conditions $h_F(Q_v) = \zeta_v$ at the corners are the linear system

$$\begin{pmatrix} 1 & Q_a^x & Q_a^y \\ 1 & Q_b^x & Q_b^y \\ 1 & Q_c^x & Q_c^y \end{pmatrix} \begin{pmatrix} c \\ m_1 \\ m_2 \end{pmatrix} = \begin{pmatrix} \zeta_a \\ \zeta_b \\ \zeta_c \end{pmatrix}.$$

So in particular, if the coordinates of Q and ζ are in $\mathbb{Q}$ or $\mathbb{Q}(\sqrt{3})$ then so are m and c .

The lifted face is the graph of th_F over its shadow, in particular injective, and two lifted faces meet over a point x common to their shadows exactly when the *gap* $h_F - h_G$ vanishes at x . The vertical distance between the two sheets over x is $t|h_F(x) - h_G(x)|$, so t scales how far apart they are but not whether they meet. If two sheets are separated at $t = 1$, they are separated for every $t > 0$.

The next lemma describes the conditions we use to prove that our push direction indeed separates our planar configuration Q . We consider all simplices closed, and the *shared simplex* of two faces is their intersection $F \cap G$ in the triangulation, a vertex or an edge or empty. It is read off the face lists and does not depend on the configuration.

For a face F and edge (p, q) of F , let $\ell(x) = \pm \det(Q_q - Q_p, x - Q_p)$, signed so that its value at the third vertex of F is positive. Equivalently $\ell(x) = \langle n, x - Q_p \rangle$ with $n = \pm J(Q_q - Q_p)$ the normal of the edge, the sign chosen such that n points into the face. We call ℓ the *edge form* of (p, q) in F . The shadow of F is exactly the set where its three edge forms are nonnegative.

The *overlap* $K = A \cap B$ of two shadows is a compact convex polygon, possibly empty and possibly degenerate, and its *corners* are its extreme points, the points of K not interior to any segment of K .

Lemma 1 (separation criterion). *Let $F \neq G$ be faces with shadows A and B , gap $g = h_F - h_G$, and overlap K , and let $t > 0$. Assume F and G*

- (a) SHARE NOTHING. *If g has one strict sign at every corner of K , then the lifted faces are disjoint.*
- (b) SHARE A SINGLE VERTEX s . *If $\sigma g = c_1 \ell_i + c_2 \ell_j$ for some sign $\sigma \in \{+1, -1\}$, some edge forms of F or of G vanishing at Q_s with independent linear parts, and some weights $c_1, c_2 > 0$, then the lifted faces meet only at the lifted vertex.*
- (c) SHARE AN EDGE (a, b) . *If the two lifts have different slopes across the shared edge, $\langle m_F - m_G, n \rangle \neq 0$ for its normal n and the linear parts m_F, m_G of the two height functions, then the lifted faces meet exactly along the lifted edge.*

Proof. The lifted faces meet over x precisely when x lies in K and $g(x) = 0$.

(a) A compact convex planar set is the convex hull of its extreme points, here the corners, and an affine function respects convex combinations, so g is bounded on K between its smallest and largest corner values. One strict sign at every corner therefore keeps g away from zero on all of K , and if K has no corners it is empty and there is nothing to check.

(b) Each edge form is nonnegative on its own shadow, hence on K , so $\sigma g \geq 0$ on K , and a zero of σg on K forces ℓ_i and ℓ_j to vanish there, both weights being strictly positive. Two affine functions with independent linear parts have exactly one common zero, and Q_s is it, both edge forms passing through it. It lies in K , being a corner of both shadows, and both heights are ζ_s there, so the two lifted faces meet exactly at the lifted vertex $(Q_s, t\zeta_s)$.

(c) The gap vanishes at Q_a and at Q_b , both heights being ζ_a and ζ_b there, hence on the whole line L through them, so $g = c \ell_{ab}$ with ℓ_{ab} the edge form of the shared edge, the affine functions vanishing on L being its multiples. Pairing the linear parts with n gives $\langle m_F - m_G, n \rangle = c |n|^2$, so the slope condition says exactly $c \neq 0$, and the zero set of g is L . A nondegenerate triangle meets the line of its own edge in that edge, so $K \cap L$ is the shadow of (a, b) , and over it both lifts agree with the lifted edge. $\square$

In (b) the combination always exists. The affine functions vanishing at Q_s form a two dimensional space and two edge forms through Q_s with independent linear parts are a basis of it, so the gap, which vanishes at Q_s as well, is a unique combination of them. Computing c_1, c_2 for given ℓ_i, ℓ_j is solving $c_1 n_i + c_2 n_j = \sigma(m_F - m_G)$ with n_i, n_j the normals of the two edge forms. Only the positivity of the two weights has to be checked. It makes σg positive on the sector between the two lines, which contains the overlap, with Q_s the only zero. And it has to be strict, since with $c_2 = 0$ the gap would vanish along the whole line of ℓ_i and the sheets would intersect where that line meets the overlap.

Figure 6 draws case (a) on real data, the pair of faces of the square base on which the smallest corner gap of Proposition 2 is attained.

A configuration places each face in space as the triangle spanned by its three vertices, and these triangles fit together into a map from T to $\mathbb{R}^3$ that is affine on each face. The configuration is *embedded* when that map is injective. The following lemma is straightforward.

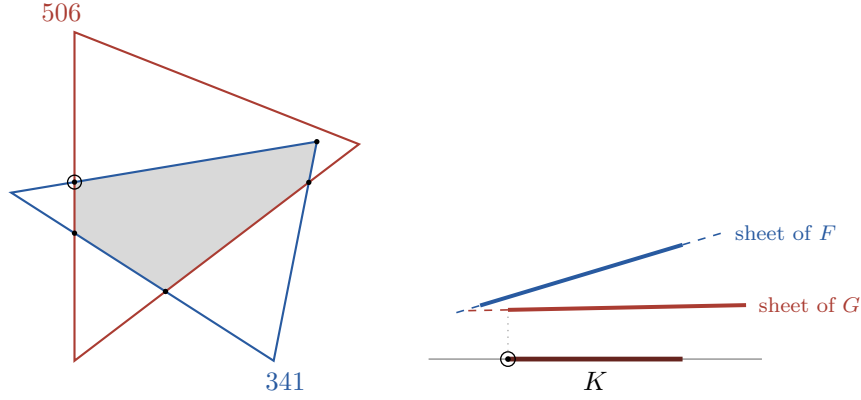


FIGURE 6. Case (a). Left, the shadows of the faces 341 and 506 of the square base, the overlap filled, its five corners dotted, and the corner circled where the smallest gap of Proposition 2 is attained. Right, a schematic profile of the same pair, each sheet solid over its own shadow and the sheets at least $0.0054t$ apart over the marked overlap.

Lemma 2 (pairwise embedding criterion). *Let q be a configuration with nondegenerate faces such that the images of every two distinct faces meet exactly in the image of their shared simplex. Then q is embedded.*

Each pair of faces of the base now has to pass its case of the criterion.

Proposition 2. *For the square base Q^0 with the lift parameter*

$$\zeta = \left(\frac{3}{13}, -\frac{3}{8}, -\frac{3}{7}, \frac{12}{13}, 1, -1, 1, 0 \right),$$

and for the hexagonal base P^0 with the lift parameter

$$\zeta' = \left(\frac{1}{2}, -1, \frac{6}{17}, -\frac{7}{34}, -\frac{33}{47}, -1, \frac{19}{41}, 1 \right),$$

the lift is embedded for every $t > 0$.

Proof. Every one of the $\binom{16}{2} = 120$ pairs of faces of each base satisfies its case of Lemma 1, in exact arithmetic in its field. Everything is rational in the printed data, so the checks are finitely many exact sign evaluations, of the gap at the corners in case (a), of the two weights from one 2×2 solve per choice in case (b), of one ratio in case (c). On the square side, twenty-four pairs share an edge, seventy-two share exactly a vertex, and twenty-four share nothing, seventeen of the last with overlapping shadows. The smallest value of $|g|$ at a corner of an overlap is $\frac{132043}{24086296}$, about 0.0055, against a diameter of about two for Q^0 , the smallest weight in case (b) is $\frac{3301075}{551669833}$, about 0.0060, and the smallest slope difference across a shared edge, in units of its edge form, is $\frac{1320430}{262864459}$, about 0.0050. On the hexagonal side, twenty-four pairs share an edge, seventy-five share exactly a vertex, and twenty-one share nothing, sixteen of the last with overlapping shadows. The smallest corner gap is

$$\frac{-25244566233401343 + 14596454895110643\sqrt{3}}{2513111807172148},$$

a positive element of $\mathbb{Q}(\sqrt{3})$, approximately 0.0148, against a diameter of about one for P^0 , the smallest weight is $\frac{997745613375 - 89808500275\sqrt{3}}{10085314779628}$, approximately 0.0835, and the smallest slope difference, in units of its edge form, is

$$\frac{71551414451157524171250 - 35233592317359027252300\sqrt{3}}{467889697181773607674453},$$

about 0.0225. The script `separation.py` of Appendix A runs all of it, and Lemmas 1 and 2 turn the checks into embeddedness, for every $t > 0$ at once. $\square$

The next lemma tells us that once a lift is embedded, so are the lifts of a whole neighborhood of the planar coordinates, at every height.

Lemma 3 (persistence of the separation). *There is, for each base, a neighborhood U of it in $(\mathbb{R}^2)^V$, the same for every $t > 0$, such that for every planar configuration $Q \in U$ and every $t > 0$, the lift of Q with heights $t\zeta$, respectively $t\zeta'$, is an embedded torus.*

Proof. Embeddedness is an open condition on a configuration: at an embedded one, each pair of faces meets exactly in the image of its shared simplex by a positive margin, the margins move continuously with the vertices, and Lemma 2 keeps every nearby configuration embedded. The lift at $t = 1$ is continuous in Q and embedded at the base by Proposition 2, so the planar configurations with embedded lift at $t = 1$ form an open neighborhood U of the base, and the invertible map $(x, y, z) \mapsto (x, y, tz)$ carries that lift to the lift at height t , so the same U serves every $t > 0$. $\square$

For where ζ and ζ' come from, see Remark 1 and Section 7.1.

6. THE CORRECTION

The lifts of Q^0 and P^0 are embedded for every $t > 0$ (Proposition 2), but they are not flat. We correct each by moving nine of the sixteen planar coordinates while the heights stay frozen.

As a reminder, the eight deficits sum to zero (Section 2), so vanishing of the first seven is enough. Adding the two equations that pin the modulus gives the map

$$\Phi = (\delta_0, \dots, \delta_6, \operatorname{Re} \tau - \operatorname{Re} \tau_0, \operatorname{Im} \tau - \operatorname{Im} \tau_0),$$

where $\tau = v_2/v_1$ is the modulus relative to the marking of Section 4, computed along the fixed unfolding tree, each loop edge read in its earlier face (Remark 3), and τ_0 is the target marked modulus, i for the square base and ρ for the hexagonal one. A zero of Φ lies on the flat locus $\mathcal{F}$, all eight deficits vanishing by the angle count of Section 2, and has modulus $\tau = \tau_0$ (Section 2.1). We read Φ as a function of nine free planar coordinates, the other seven planar coordinates frozen at their planar values and the eight heights frozen at their lifted values. For the square base the nine free coordinates are

$$Q_0^x, \quad Q_0^y, \quad Q_1^x, \quad Q_1^y, \quad Q_2^y, \quad Q_3^x, \quad Q_4^x, \quad Q_4^y, \quad Q_5^x,$$

and for the hexagonal base they are

$$P_0^x, \quad P_0^y, \quad P_1^x, \quad P_1^y, \quad P_2^x, \quad P_2^y, \quad P_3^x, \quad P_3^y, \quad P_4^x,$$

both selected in Proposition 3.

Remark 2. The implicit function theorem needs Φ continuously differentiable in an open neighborhood around the base, and not only along the flat locus through it. It is, because Φ is assembled from maps that are smooth away from degenerations. The unfolding places the third vertex of each next face at the two prescribed distances from the ends of the shared edge and on the prescribed side, smoothly as long as the radicand of the doubled area stays positive, which is to say as long as the face stays nondegenerate. The modulus reads the two periods off the net and forms $\tau = v_2/v_1$, smooth while v_1 is nonzero, and each deficit is 2π minus a sum of triangle angles, smooth on nondegenerate triangles. So Φ is smooth wherever the sixteen faces are nondegenerate and v_1 does not vanish, both of which hold at each base, by Section 3 and by $\det(v_1, v_2) > 0$ from Section 4.

Remark 3. Off the flat locus the modulus depends on two choices, the unfolding tree and the face in which each marking edge is read. Each tree and each such choice of faces define their own smooth extension of τ to configurations that are not flat, and all of them agree on the flat locus. We use the extension belonging to the breadth first tree and the earlier face of each loop edge, the tree fixed in Section 4, and the Jacobian below is its Jacobian.

The Jacobian of Φ at a planar base lies in the field of the base, even though Φ itself does not. At the corner A of a triangle ABC write $\alpha = |AB|^2$, $\beta = |AC|^2$, $\gamma = |BC|^2$, so that $\langle AB, AC \rangle = \frac{1}{2}(\alpha + \beta - \gamma)$, and let $S = 2 \text{Area}(ABC)$. The pair $(\langle AB, AC \rangle, S)$ is $|AB||AC|(\cos \theta_A, \sin \theta_A)$, so the angle at A is the argument of this plane vector. The differential of an argument is the winding form $d \arg(x, y) = (x dy - y dx)/(x^2 + y^2)$, obtained by differentiating $\arctan(y/x)$, whose branches differ from $\arg$ by constants, and the squared length of the pair is $\langle AB, AC \rangle^2 + S^2 = \alpha\beta$, which is Lagrange's identity, so

$$d\theta_A = \frac{\langle AB, AC \rangle dS - S d\langle AB, AC \rangle}{\alpha \beta},$$

with everything on the right rational in the squared edge lengths and in S . At a planar configuration S is the orientation sign of the face times its planar cross product, by Lagrange's identity as in Section 4, and dS along planar directions is that sign times the derivative of the cross product, so $d\theta_A$ lies in the field at the base and no square root is taken.

Proposition 3. *At each planar base the 9×16 Jacobian of Φ with respect to the planar coordinates has entries in the field of the base and exact rank 9. Gaussian elimination selects the nine free coordinates listed above, and the resulting 9×9 matrix M is invertible.*

Proof. The script `jacobian.py` of Appendix A, in exact arithmetic on the printed data, over $\mathbb{Q}$ on the square side and over $\mathbb{Q}(\sqrt{3})$ on the hexagonal one. It rebuilds the unfolding of Section 4 from each base, assembles the $9 \times 16 = 144$ entries from the angle derivative formula above, carrying one gradient alongside every intermediate quantity by the chain rule, one arithmetic operation at a time. On the square side

$$\det M = \frac{4243180825891307520000}{4349139164624211581} \neq 0,$$

and on the hexagonal side $\det M$ is a number $a + b\sqrt{3}$ whose two rational parts have 89 digit numerators over a common 85 digit denominator, printed in full by the script, both parts negative, so the determinant is nonzero, approximately -7.328×10^4 . $\square$

Two structural facts finish the construction, the same on each base. The heights enter Φ only through the squared edge lengths, and the squared edge length of (a, b) at heights $t\zeta$ is

$$|Q_a - Q_b|^2 + t^2(\zeta_a - \zeta_b)^2,$$

so Φ depends on t only through $s = t^2$. Write $\Psi(x, s)$ for the resulting map of the nine free coordinates and that one parameter, so that Φ at heights $t\zeta$ is $\Psi(\cdot, t^2)$. It is defined for s of either sign, the unfolding consuming only the twenty-four numbers $|Q_a - Q_b|^2 + s(\zeta_a - \zeta_b)^2$, which keep every radicand positive and v_1 nonzero near the base. Then $\Psi(x^0, 0) = 0$, by Proposition 1, and Ψ is smooth near $(x^0, 0)$ by Remark 2, with $\partial_x \Psi(x^0, 0) = M$ invertible by Proposition 3.

Lemma 4 (correction lemma). *On each base there are $\varepsilon > 0$ and a continuous map $s \mapsto x(s)$ on $[0, \varepsilon)$ with $x(0) = x^0$ and $\Psi(x(s), s) = 0$. In particular, for $0 < t < \sqrt{\varepsilon}$ the configuration $q(t)$ whose nine free planar coordinates are $x(t^2)$, whose other seven planar coordinates are those of the base, and whose heights are $t\zeta$, respectively $t\zeta'$, has all eight cone angles equal to 2π and modulus exactly τ_0 relative to the marking.*

Proof. The implicit function theorem applied to Ψ at $(x^0, 0)$, where $\partial_x \Psi$ is the invertible matrix M , gives $\varepsilon > 0$ and a continuously differentiable map $x(s)$ on $(-\varepsilon, \varepsilon)$ with $x(0) = x^0$ and $\Psi(x(s), s) = 0$. The last sentence is the substitution $s = t^2$ together with the deficit sum, which turns the seven vanishing deficits into eight. $\square$

Nothing quantitative about the curve $s \mapsto x(s)$ is needed, the argument below consuming only its continuity at $s = 0$. A marking is a pair of loops on T and does not move when the configuration does, so its classes form a basis of $H_1(T; \mathbb{Z})$ all along the curve, by the intersection argument of Section 4, and the modulus τ_0 of Lemma 4 is therefore the reduced modulus.

Proof of Theorems 1 and 2. On each base, let $Q(s)$ be the planar configuration whose nine free coordinates are the curve $x(s)$ of Lemma 4 and whose other seven are those of the base. By continuity $Q(s)$ stays in the neighborhood U of Lemma 3 for all small s , so for small $t > 0$ the configuration $q(t)$ with planar part $Q(t^2)$ and the lifted heights is an embedded torus, and it is exactly flat of reduced modulus i , respectively ρ , by Lemma 4 and the paragraph closing Section 6. No torus triangulation has fewer than seven vertices (Section 2), and no paper torus has seven [10], so eight is minimal. The nine components of $x(t^2)$, less their base values, are the continuous functions of the statements, vanishing at $t = 0$. $\square$

The proof gives more than the two tori.

Theorem 3. *On each of the two triangulations there is an open neighborhood W of the printed folded base in the set $\mathcal{C}^\circ$ of configurations with nondegenerate faces, with the following three properties. The flat locus $\mathcal{F} \cap W$, the configurations in W with all eight cone angles 2π , is a smooth manifold of dimension seventeen. The modulus is a submersion on it, so each of its nonempty fibers is a smooth manifold of dimension fifteen. And the fiber over the target, i on T and ρ on T' , contains the configurations $q(t)$ of the proofs of Theorems 1 and 2 for all sufficiently small $t > 0$, and these are embedded.*

Proof. The map Φ is continuously differentiable on the open set of configurations with nondegenerate faces and $v_1 \neq 0$ (Remark 2), so the determinant of the 9×9 minor of Φ in the nine free coordinates of Proposition 3 is continuous there, and it is nonzero at the base. Take W to be the open neighborhood of the base where it stays nonzero and where $\det(v_1, v_2)$, positive at the base (Section 4), stays positive, so that τ maps $\mathcal{F} \cap W$ into $\mathbb{H}$. The 9×24 Jacobian of Φ then has rank nine at every point of W .

The seven deficit rows of that Jacobian are rows of a rank-nine matrix, so they are linearly independent at every point of W . Zero is therefore a regular value of $(\delta_0, \dots, \delta_6)$ on W , its zero set $\mathcal{F} \cap W$ is a smooth manifold of dimension $24 - 7 = 17$, and the tangent space at each of its points is the kernel of the seven deficit differentials. On $\mathcal{F}$ the fixed-tree extension is the modulus of Section 2.1 (Section 4), so the differential of τ on that tangent space is the restriction of the two modulus rows. Suppose τ failed to be a submersion at a point of $\mathcal{F} \cap W$. Some nontrivial combination of the two modulus rows would then be orthogonal to the tangent space there, the common kernel of the seven deficit rows. A vector orthogonal to that kernel lies in the span of the seven, the row space of a matrix being the orthogonal complement of its null space. So the nine rows would be dependent, contradicting rank nine. So τ is a submersion on $\mathcal{F} \cap W$, and a nonempty fiber is a smooth manifold of dimension $17 - 2 = 15$.

The configurations $q(t)$ converge to the base as $t \rightarrow 0$, their planar part being the continuous curve $Q(t^2)$ with $Q(0)$ the base and their heights being the lifted ones, so they lie in W for all small $t > 0$. They are zeros of Φ (Lemma 4), which puts them in the fiber over the target, and they are embedded for small t by the proof of Theorems 1 and 2. $\square$

The two rays that near universality excludes both start at an orbifold point, and the submersion covers an initial segment in each.

Corollary 1. *There is an $\varepsilon > 0$ such that every rectangular torus $\hat{\tau} = iY$ with $1 \leq Y < 1 + \varepsilon$ and every rhombic torus $\hat{\tau} = \frac{1}{2} + iy$ with $\frac{\sqrt{3}}{2} \leq y < \frac{\sqrt{3}}{2} + \varepsilon$ is realized by an 8-vertex paper torus. Eight-vertex paper tori therefore realize nontrivial initial segments of both rays left open by [11].*

Proof. Fix a small $t > 0$ and let $p = q(t)$ on Q^0 , an embedded point of the fiber of τ over i (Theorem 3). The set $\mathcal{E}$ of embedded configurations is open (Section 2.1), so p has an open neighborhood $N \subset \mathcal{E} \cap W$, and $\mathcal{F} \cap N$ is an open neighborhood of p in the manifold $\mathcal{F} \cap W$. A submersion is an open map, so $\tau(\mathcal{F} \cap N)$ contains a ball B around i in $\mathbb{H}$, and every point of $\mathcal{F} \cap N$ is flat and embedded, a paper torus of marked modulus τ . The rhombic segment comes from $p = q(t)$ on P^0 the same way, and the smaller of the two ball radii serves as ε . The marked moduli in the statement are their own reduced representatives, iY with $Y \geq 1$ lying in $\mathcal{D}$ and $\frac{1}{2} + iy$ with $y \geq \frac{\sqrt{3}}{2}$ as well, by $|\tau|^2 = \frac{1}{4} + y^2 \geq 1$ and the convention $\operatorname{Re} \hat{\tau} \geq 0$. $\square$

7. DISCUSSION

7.1. Where the planar configurations come from. The folded configurations of Table 1 were found, not guessed. In a larger program on the moduli left open by [11], we first produced tori of the excluded moduli numerically, flat to machine precision and embedded by positive margins. A flow then drove each of them toward planarity, shrinking the volume while barriers kept the face separation positive and a projection kept flatness and the pinned modulus. Doyle and Schwartz lose their tori along the two rays, because the vertices of their collapsing families collide in the limit. The flow instead holds the modulus fixed on the ray and we read off what the collapse to the plane looks like there.

The limits that arrived were different in kind. The vertices did not sit in general position. They fell into incidences, collinearities on the rectangular side and concyclicities on the rhombic one, persisting along each ray. The incidences are worth recording, exact on the printed coordinates and drawn in Figure 7. On the square base the vertices 0, 2, 4 are collinear, the straight crease, the vertices 0, 4, 7, 6 form a rectangle, and 5 lies on its circumscribed circle. The hexagonal base carries concyclicities instead, the five vertices 0, 2, 4, 5, 6 on one circle and the quadruples 0, 1, 3, 6 and 3, 4, 5, 7 on two more.

Closing the incidences, imposing them as exact equations and solving, produced the two four-parameter rational families of Remark 1, one sweeping the rectangular line and one the rhombic line, and the directions ζ and ζ' came with them, carried by the families on the sections through their corners (Section 5). The two bases are those families at their corners $Y = 1$ and $h = \frac{\sqrt{3}}{2}$. Everything in this subsection is floating point until the incidences close, and none of it enters the proof, which consumes Table 1 and the two directions, nothing else (Remark 1).

Taken with the sequel, which opens both lines, and with [11], which supplies every modulus off the two excluded rays, the two corners proved here complete the realization of every flat torus by a vertex-minimal paper torus.

7.2. Open questions. The theorem is an existence statement with no explicit t . We also have no inflated representative of either modulus with exact coordinates. The tori of Appendix B are numerical. Two more questions concern the triangulations. Of the seven 8-vertex torus triangulations, the one with degree sequence (4, 5, 5, 6, 7, 7, 7, 7) resisted every numerical search (Appendix B), and we do not know whether it carries any flat torus at any modulus. And near universality is realized on the valence-regular T alone, while the hexagonal torus is realized here on T' , so we do not know whether a single triangulation carries every flat torus. A last question

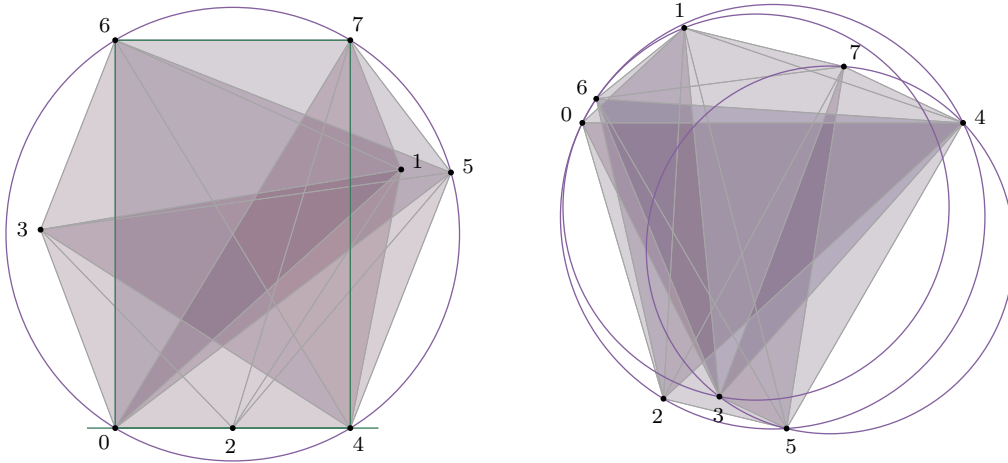


FIGURE 7. The defining incidences of the two folded bases, drawn on the bases themselves. On the left the line through the vertices 0, 2, 4 and the rectangle 0, 4, 7, 6 are green and the circumscribed circle of that rectangle is purple, with the vertex 5 on it. On the right the three circles of the hexagonal base are purple, the first through 0, 2, 4, 5, 6 and the other two through 0, 1, 3, 6 and through 3, 4, 5, 7. We compute each center and radius exactly from the printed coordinates and check each vertex on its circle before drawing.

is hands-on. We would like to say that Theorems 1 and 2 mean exactly that the square and hexagonal tori can be folded from a flat sheet of paper along the edges of their triangulations, but already the precise question is unclear to us, an isotopy moving each face by isometries and keeping every edge straight, as in the motion of a mechanical linkage, the same with new creases allowed as in origami, a mere isotopy of embeddings, or something else, and we settle no version of it. Both inflated tori of Appendix B have been folded from their nets by hand.

APPENDIX A. VERIFICATION

Every claim the paper makes about the two bases is checked by short Python scripts using exact arithmetic on the data of Table 1, plain fractions on the square side and pairs of fractions representing $a + b\sqrt{3}$ on the hexagonal one. No floating point number enters any accepted claim, and each individual check is small enough to redo by hand or in a computer algebra system.

script	checks	rows in the text
<code>combinatorics.py</code>	each face list is a triangulated torus	Section 3
<code>jacobian.py</code>	face signs, deck data, holonomies, τ , $\det M \neq 0$	Props. 1, 3
<code>separation.py</code>	all 120 face pairs, by shared simplex	Prop. 2
<code>qsqrt3.py</code>	the ordered field $\mathbb{Q}(\sqrt{3})$	Section 3
<code>hexbase.py</code>	the hexagonal base from its closed forms	Table 1

The scripts live in the folder `code/` [TODO: github link. What is best/common here?] and run from the single command `python3 run.py`, one log per script. The whole run, for both tori, takes under five seconds. Each of them reads the printed data from one frozen file. The combinatorics script tests five conditions on the two face lists: distinct faces, every directed edge traversed by exactly one face cycle, a single cycle of faces around each vertex, connectedness, and $|F| = 2|V|$. The first four make each list a connected closed oriented surface, and the count

$8 - 24 + 16 = 0$ then makes it a torus. The Jacobian script also checks each of the thirty-two printed coordinates against the closed forms of the families behind Remark 1, and rebuilds the unfolding. It also checks that the eight vertex holonomies of each base are the identity, redundant given the nine gluing checks by the argument of Section 4, and kept as a sanity check. One control runs alongside, not part of any proof: the Jacobian agrees, entry by entry and to within 2×10^{-9} , with central differences of an independent floating point evaluation of Φ .

APPENDIX B. TWO INFLATED TORI

The tori of Theorems 1 and 2 are nearly planar, and no scissors will reproduce them. This appendix records an inflated torus for each, found by a numerical search on the same two triangulations. A rough embedded candidate near the target modulus was refined by alternating Gauss–Newton projection onto the numerically flat, pinned-modulus locus with gradient descent of an embedding energy along it, until the residual reached 10^{-15} . The results are the configurations of Tables 4 and 5, drawn in Figure 1. Read as exact rational numbers, both are exactly embedded, verified by a separating-axis reduction in exact rational arithmetic, and neither carries a spatial symmetry, no isometry of $\mathbb{R}^3$ permuting the vertices, verified exactly as well. The scripts and the coordinates are at

<https://github.com/FabianLander/square-hex-paper-tori>

Neither printed configuration is exactly flat. The cone angles equal 2π to within 3×10^{-15} and the marked moduli match i and ρ to fifteen digits. The repository does contain more: a machine-verified Krawczyk certificate [15] proving that within 10^{-9} of each printed configuration, coordinate by coordinate, there is an exactly flat one of marked modulus exactly i , respectively exactly ρ . That proof rests on validated interval arithmetic, which this paper deliberately avoids, and the theorems here stand on the folded bases alone.

The search also looked past the two triangulations used here. Of the seven 8-vertex torus triangulations, the square modulus was realized numerically on five. One of the remaining two has a vertex of valence 3, where the three angles sum to 2π by flatness, and three unit vectors whose pairwise angles sum to 2π are coplanar, so the star is flat and erasing the vertex leaves a 7-vertex paper torus, impossible by [10]. The last, with degree sequence $(4, 5, 5, 6, 7, 7, 7, 7)$, resisted every search, we found no flat torus on it at any modulus, and we conjecture that none exists.

Figures 8 to 11 are printable nets, two per torus, each carrying the whole surface as one connected sheet. Solid lines are cuts. A green dash-dotted crease is a mountain and a purple dashed one a valley, the dash patterns carrying the distinction without the color, and each crease is labeled M or V together with the angle between its two faces, measured on the printed side, which is the side that ends up outside. A letter pairs a glue tab with the edge it is glued behind. The fold angles are the dihedral angles of the printed coordinates, rounded, and they only indicate roughly how far to fold. No measuring is involved, the creases settling into place as the seams close. The two nets of a torus differ only in which fifteen or more of the twenty-four edges stay joined, and there is a trade-off. The spread net leaves nine seams to glue but puts every vertex on the cut boundary, where the paper can be reached from outside. The compact net keeps more edges joined, because a fan of faces closing up around a flat vertex meets itself with no gap and so needs no glue there, leaving only five seams for the square torus and six for the hexagonal one. It also closes up about twice as accurately, the folds composing along shorter chains, at the price of four interior vertices, respectively three, that have to be creased in the middle of the sheet, which is hard to do by hand. The red and blue bands mark the two loops whose holonomies are the periods v_1 and v_2 , the loops of Section 4, so a sheet also shows where its modulus is read off. The objects are genuinely awkward, nine of the square torus’s twenty-four folds closing to within fifteen degrees of flat on flat. We can report that both tori do fold up in practice.

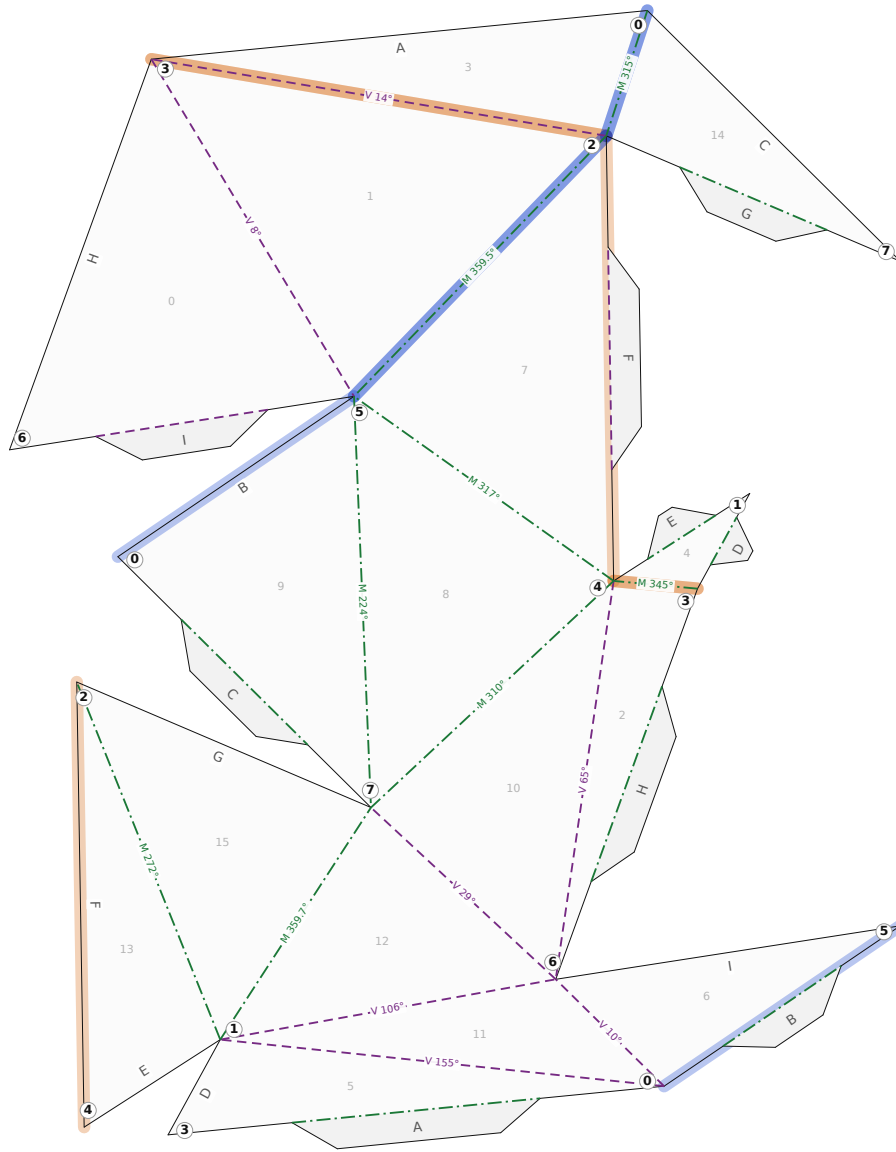


FIGURE 8. The cut-out net of the inflated square torus, the spread version: fifteen creases, nine glued seams, no interior vertex.

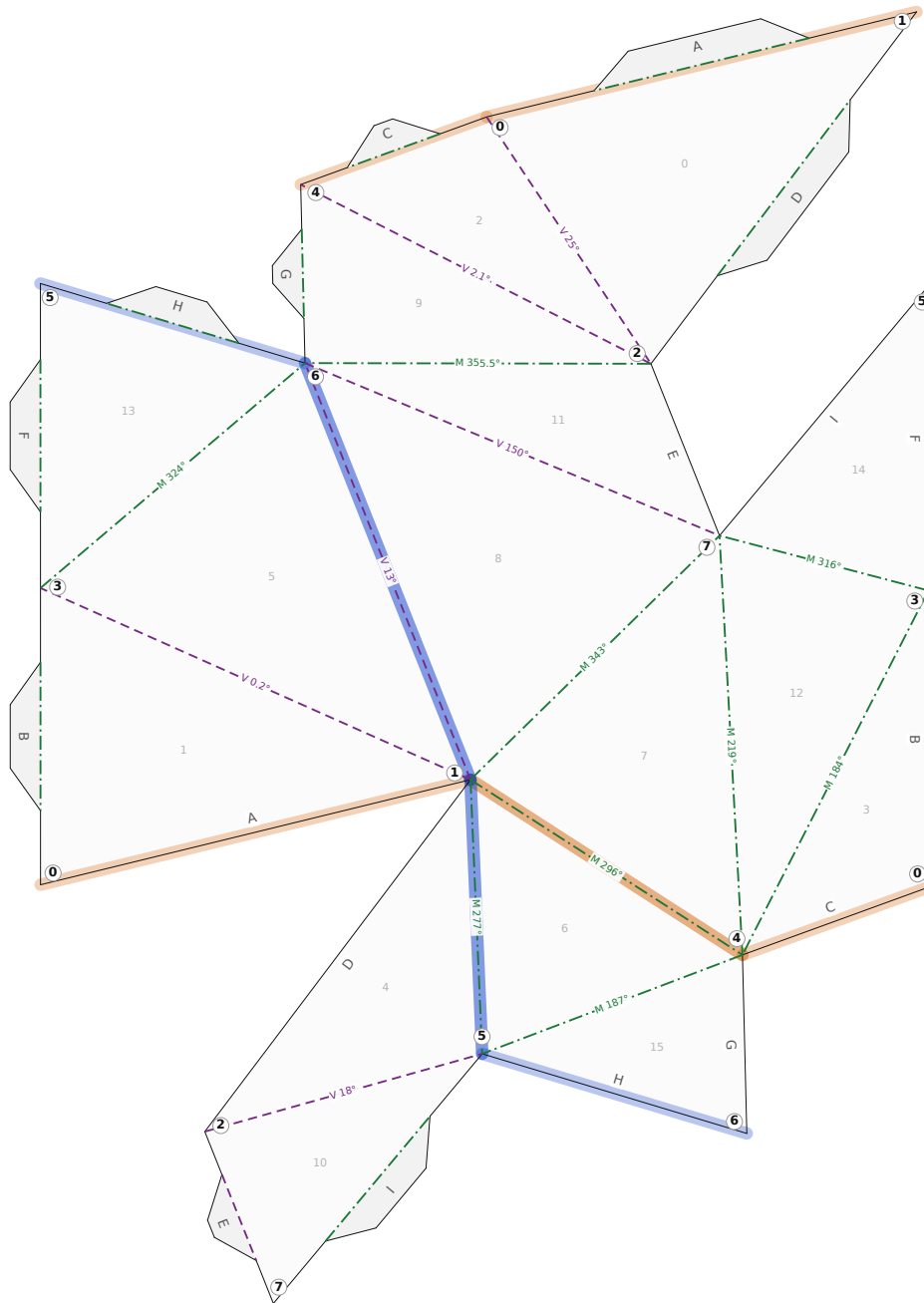


FIGURE 9. The cut-out net of the inflated hexagonal torus, the spread version: fifteen creases and nine glued seams, again with no interior vertex.

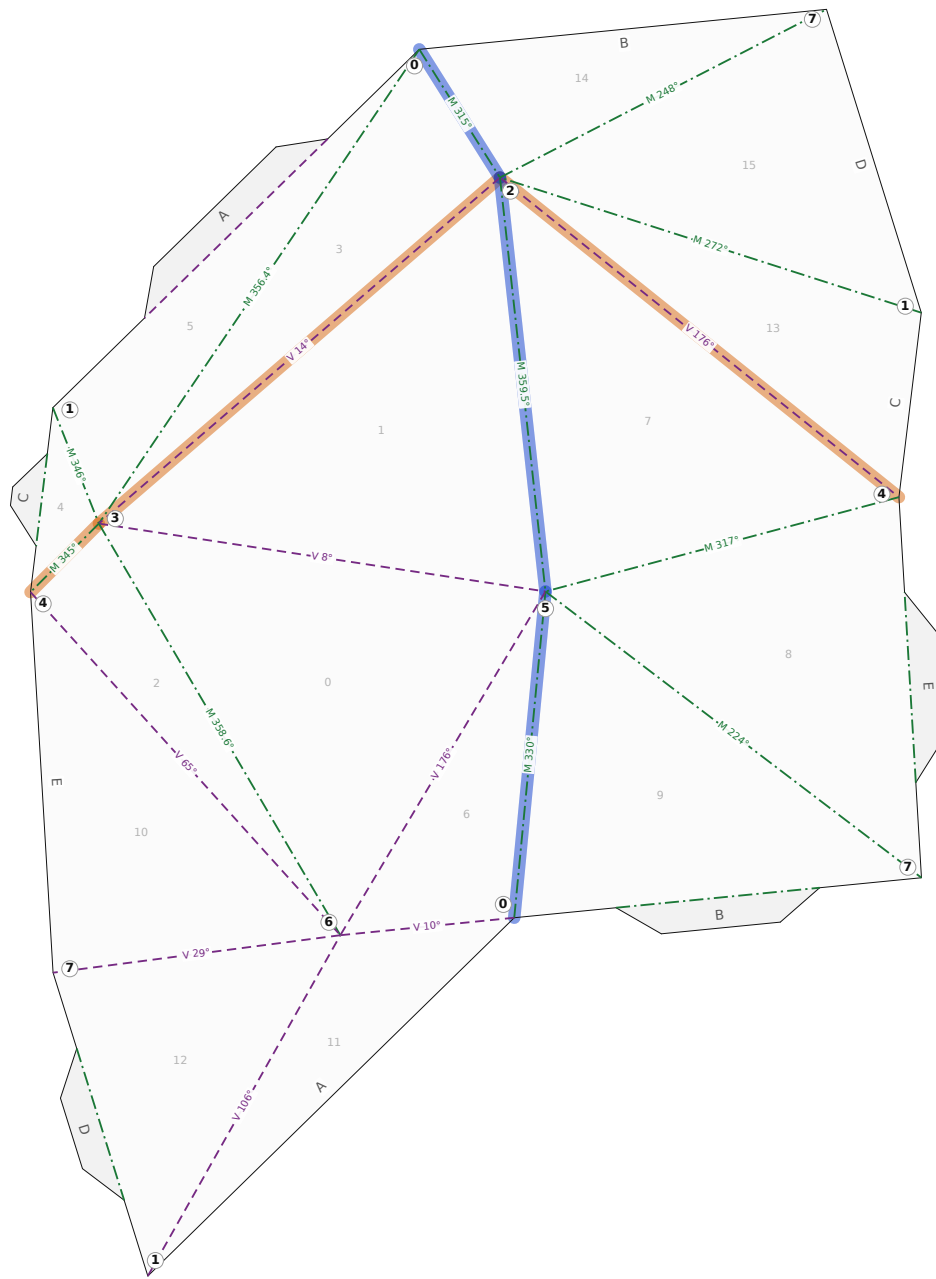


FIGURE 10. The inflated square torus cut open the other way: nineteen creases, only five glued seams, and it folds up about twice as accurately as Figure 8. Four of its vertices lie inside the sheet rather than on the cut boundary, which makes it the harder of the two to fold by hand.

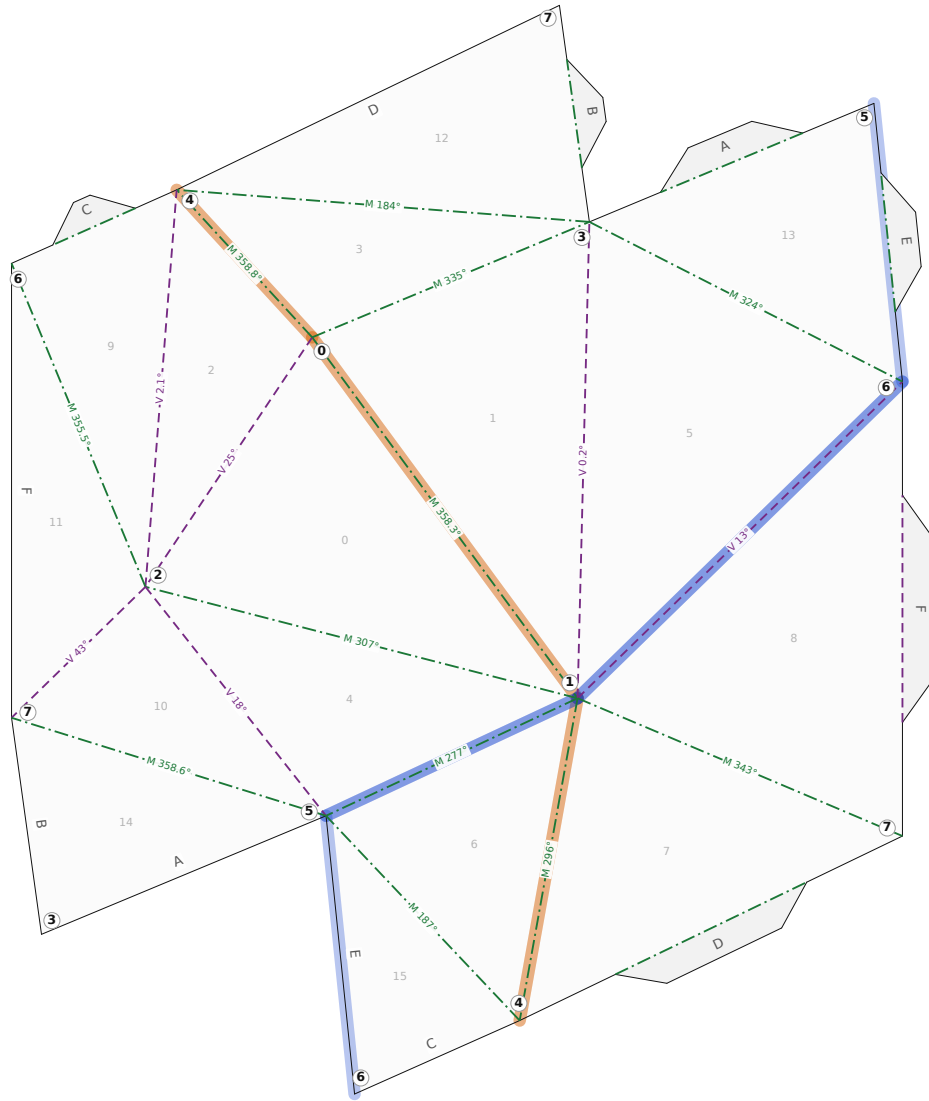


FIGURE 11. The inflated hexagonal torus along its own development: eighteen creases, six glued seams, three interior vertices.

v	x_v	y_v	z_v
0	-0.32228466828574437	0.4637721192185657	-0.74323457006951088
1	0.2324200435776736	-0.46217058403057881	0.24599012355582775
2	-0.14382327673249617	0.079722048364084933	-0.83230338087592548
3	0.22779161128071213	-0.32944176958982457	0.57668754870786809
4	0.17876493830658916	-0.057582559083758984	0.58547474269171451
5	-0.12429063916040699	0.81747961438065686	0.10086387909811872
6	-0.24593729785582902	-0.023388654255220555	-0.66463839618166654
7	-0.62529227417585398	-0.42715999499773932	-0.049686468538500325

TABLE 4. The inflated square torus, the eight vertex positions to full double precision, from the data file `square-i-type7.csv` of the repository above.

v	x_v	y_v	z_v
0	-0.0052919576658824819	0.25189213380754671	-0.19636317590069088
1	-0.10330037049737666	0.097954858850342955	0.23022989757571455
2	0.088435194516004872	-0.042881936616869155	-0.16602959156361255
3	0.10713507949378637	-0.03807120945413147	-0.19492235829557528
4	-0.15821722275466177	0.26928131058897076	-0.057266978045909966
5	0.099801133842719214	0.17571318949877185	0.042053203465396087
6	-0.026706140635424078	0.3003490487315576	-0.18681063266426795
7	-0.004032608759816151	-0.14181859043688222	-0.02715724967818926

TABLE 5. The inflated hexagonal torus, from the data file `rho-type3.csv`.

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